MATTHEW POOL

Software Engineer | Product Designer

UI/UX | Native Mobile | Full-Stack Web | AI/ML | B2B/B2C

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matthew-pool.github.io

PROFESSIONAL SUMMARY

Software Engineer and Product Designer with a B.S. in Computer Science (3.99 GPA) blending technical knowledge with a business background including 5 years of (top-ranked) outbound B2B/B2C sales for Fortune 500 clients and 10+ years of combined customer service and technical support experience. I build things end-to-end - from design to production deployments - and can even close the sale! I excel at clear communication and finding outside-of-the-box solutions that are robust, accessible, and always focused on the user experience.

EDUCATION

Bachelor of Science in Computer Science

Southern New Hampshire University (Manchester, NH) • Sep 2021 - Dec 2024

Summa Cum Laude • ***3.99 GPA*** • **President's List (2023-2024)**

Certificate in Data Analysis • *uCertify (2024)*

Associate of Arts in General Studies

Tyler Junior College (Tyler, TX) • Aug 2003 - May 2005

TECHNICAL SKILLS

🎨 Product Design

UI/UX Design, Figma, Adobe Creative Suite, CAD, SketchUp, Responsive Design, User Flows, Information Architecture

💻 Programming Languages

Python, JavaScript, TypeScript, Kotlin, Java, C/C++, C#, SQL

🎮 Gaming & Graphics

Unity, OpenGL/GLSL, Reinforcement Learning (RL), Deep Q-Networks (DQN)

📱 Mobile Development

Android SDK, Jetpack Compose, Material Design 3, MVVM Architecture, Coroutines, Hilt, Firebase, REST APIs

🌐 Web Technologies

React, Angular, Node.js, Express.js, LoopBack API, RxJS, HTML5, CSS3, Vite, MVC, Bootstrap, Tailwind CSS, Handlebars, Spring Boot

🗄️ Databases

MySQL, MongoDB, Mongoose (ODM), AWS DynamoDB, Firebase Firestore

🤖 Data Science & Machine Learning

TensorFlow, scikit-learn, Pandas, NumPy, Matplotlib, Deep Neural Networks (DNN), Convolutional Neural Networks (CNN)

🧪 Testing & QA

Postman, pytest, JUnit, MockK, Mockito, Espresso, Jacoco, Jest, Selenium

⚙️ DevOps & Security

Docker/Docker Compose, AWS, CI/CD, Git/GitHub Actions, JWT, OWASP, Snyk, TLS/SSL, RSA/AES, Gradle, Maven

TECHNICAL PROJECTS

Full-Stack Travel Platform

Angular, TypeScript, LoopBack API, Docker, AWS DynamoDB

- Engineered an Angular/TypeScript SPA utilizing the LoopBack API framework, orchestrating a Docker-containerized database migration from MongoDB to AWS DynamoDB.

The Existence Paradox

React, TypeScript, Tailwind CSS, Vite

- Developed an interactive, responsive web application featuring dynamic state management and complex decision-tree navigation.

Financial Security Audit & Remediation

Java, Spring Boot, OWASP, AES/SHA Cryptography

- Refactored a legacy RESTful API to implement AES-256 encryption at rest, mitigating injection risks and ensuring OWASP compliance.

Austin Animal Center Dashboard

Python, MongoDB, Dash, Pandas

- Architected an interactive MVC data dashboard with custom Python CRUD modules, Dash data visualization, and geolocation mapping.

Machine Learning Models

Python, TensorFlow, Scikit-Learn, NumPy

- Trained and optimized Convolutional Neural Networks (CNN) for image recognition and Deep Q-Learning Networks (DQN) for pathfinding optimization.

Super Mario Bros. 3D Engine

C++, OpenGL, GLSL, GLM

- Recreated classic 3D scenes utilizing custom shaders, manual mesh generation, and optimized rendering pipelines with a Phong lighting model.

Product Design & UX

Figma, Wireframes, UI/UX Design

- Designed end-to-end user experiences for mobile and web applications, producing extensive user personas, low-fidelity wireframes, and high-fidelity interactive mockups.

Check out these projects and others at
matthew-pool.github.io

PROFESSIONAL EXPERIENCE

Lead Software Engineer & Product Designer

Jan 2025 – Present

AppBuddy (Independent Venture)

- **Product Strategy & Execution:** Architected and launched Flick, a production-ready native Android video streaming platform. Built end-to-end, beginning with UI/UX prototyping in Figma through to Google Play Store deployment.
- **Android Architecture:** Engineered a scalable native Android application using Kotlin, Jetpack Compose (Material Design 3), clean MVVM architecture, Hilt dependency injection, and offline-first data synchronization.
- **ML-Driven Personalization:** Designed and deployed a proprietary recommendation engine utilizing vectorization and cosine similarity to calculate the distance between user profiles and available content. Implemented an "exploration vs. exploitation" algorithm to dynamically rank video feeds while preventing content echo chambers.
- **Infrastructure Optimization:** Engineered a cost-efficient backend strategy by integrating Retrofit with the Archive.org API, leveraging public domain assets to reduce content delivery costs to near-zero.
- **DevOps & Quality Assurance:** Built a robust CI/CD pipeline via GitHub Actions, automating unit (JUnit/MockK) and UI tests (Espresso) to ensure virtually crash-free sessions in production.

Outbound Sales & Account Management (2004–2009)

Recurring Top Performer

Sales Representative • T.C.I.M. Services (B2B/B2C Outbound Sales)

Texas State Insurance License obtained independently within the branch.

- Conducted high-volume outbound sales calls on behalf of Fortune 500 clients including American Express, AT&T, and Citibank.
- Consistently ranked among the top performers in a rolling pool of 100+ representatives, earning "Top Performer of the Month" across multiple campaigns.
- **Training & Process Improvement:** Developed call scripts adopted across the team; trained junior sales representatives on objection handling and closing techniques.
- **Retention & Account Management:** Exceeded KPIs across acquisition, upsell, and retention metrics through effective communication and disciplined objection handling.

Technical Support, IT Operations & Inbound Sales (2011–2013)

Technical Support Specialist & Desktop Support • Sutherland / DirecTV / Frontier Communications

- **Technical Communication:** Translated complex hardware, software, VoIP, and ISP issues into clear guidance, communicating technical concepts to diverse audiences.
- **Remote Diagnostics & Documentation:** Diagnosed and resolved complex issues via remote desktop tools; documented resolutions in CRM knowledge bases to improve team-wide efficiency.
- **Inbound Sales:** Identified upsell and upgrade opportunities during support interactions, consistently meeting or exceeding sales and satisfaction targets.

Shift Management & Customer Service (1997–2003 / 2009–2010) Management & Ops

Shift Manager & Retail Associate • Kmart / Fast Food & Customer Service (East TX)

- **Team Leadership & Operations:** Directed workflow, delegated tasks, and trained new hires across fast-paced retail and food service environments.
- **Autonomy & Trust:** Regularly trusted to close stores independently at night, maintaining full financial accountability for cash registers and daily deposits.
- Acted as the direct liaison between floor staff and upper management, successfully resolving escalated customer issues.

Career Break & Independent Projects (2014–2021)

Construction & Family Management • Winnsboro, TX

- **Construction & Engineering:** Designed and constructed a residential structure (2-room tiny home) from raw materials; independently managed resource allocation, structural planning, and electrical/finish work.
- **Family Management:** Planned career break to serve as primary caregiver, manage household operations for a family of four, and facilitate spouse's successful return to college.